

2.1.1. Theoretical background and theoretical considerations

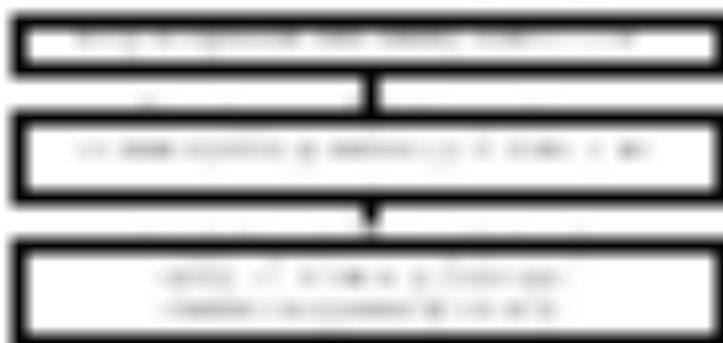
The first theoretical consideration is related to the question of the applicability of the model to the system of interest. The model is based on the assumption that the system is a linear system, which means that the output is proportional to the input. This assumption is valid for many systems, but it may not be valid for all systems. For example, the model may not be applicable to systems that exhibit non-linear behavior, such as systems with saturation or hysteresis. Therefore, it is important to check the applicability of the model to the system of interest before using it.

The second theoretical consideration is related to the question of the model's accuracy.

$$y(t) = \int_0^t x(\tau) h(t-\tau) d\tau$$

The model's accuracy depends on the quality of the data used to estimate the model's parameters.

The third theoretical consideration is related to the question of the model's robustness. The model is robust to small changes in the input, but it may not be robust to large changes in the input. For example, the model may not be able to handle inputs that are outside the range of the data used to estimate the model's parameters. Therefore, it is important to check the robustness of the model to the inputs of interest before using it.



The fourth theoretical consideration is related to the question of the model's stability. The model is stable if the impulse response $h(t)$ is bounded. This means that the output signal $y(t)$ will be bounded if the input signal $x(t)$ is bounded. The model is unstable if the impulse response $h(t)$ is unbounded. This means that the output signal $y(t)$ will be unbounded if the input signal $x(t)$ is bounded. Therefore, it is important to check the stability of the model before using it.

The fifth theoretical consideration is related to the question of the model's identifiability. The model is identifiable if the parameters can be estimated uniquely from the data. This means that the model's parameters can be determined from the data. The model is not identifiable if the parameters cannot be estimated uniquely from the data. Therefore, it is important to check the identifiability of the model before using it.

$$y(t) = \int_0^t x(\tau) h(t-\tau) d\tau$$

The sixth theoretical consideration is related to the question of the model's predictability. The model is predictable if the output signal $y(t)$ can be predicted from the input signal $x(t)$. This means that the model's output can be determined from the model's input. The model is not predictable if the output signal $y(t)$ cannot be predicted from the input signal $x(t)$. Therefore, it is important to check the predictability of the model before using it.

